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# ChatGPT as an assistant for digital scholarly editing

*ChatGPT*, OpenAI (ed.), 2022. https://openai.com/ (Last Accessed: 01.02.2026). Reviewed by Martina Scholger (University of Graz), martina.scholger@uni-graz.at and Roman Bleier (University of Graz), roman.bleier@uni-graz.at.

**Abstract**

Large language models such as OpenAI's ChatGPT have rapidly become prominent tools in academic contexts, including digital scholarly editing. Although not originally designed for editorial practice, they are increasingly integrated into experimental editorial workflows. This review asseses its potential to support key editorial tasks and surveys recent literature on their use in digital scholarly editing. Drawing on published case studies and our own experiments, we analyze the benefits and limitations of employing ChatGPT at different stages of creating and analyzing digital scholarly editions, including transcription, TEI encoding, and named-entity recognition. At the same time, the review addresses significant concerns. As a commercial and largely closed system, ChatGPT operates as a 'black box' that allows only indirect control through prompting, raising challenges for transparency and reproducibility in academic contexts. Issues of model bias, data governance, and ecological impact must be carefully considered when integrating such tools into research workflows. ChatGPT should therefore be adopted cautiously and critically. Nevertheless, its capacity to lower technical barriers and to consolidate tasks that previously required multiple tools and advanced programming skills makes it, or similar artificial intelligence tools, a potentially valuable assistant in digital editing workflows.

# Introduction

1 In the rapidly evolving landscape of the Digital Humanities, large language models (LLMs)[1] have become prominent tools for numerous tasks, including text analysis, classification, and language translation. This review examines ChatGPT's capabilities as an assisting tool for digital scholarly editing tasks. ChatGPT was selected due to its growing media presence and widespread adoption within the humanities community in recent years, including institutional deployments at universities such as the University of Graz, where the authors are affiliated.[2]

2 Introduced in November 2022 (OpenAI 2024), the chatbot ChatGPT is a generative artificial intelligence (AI) system developed by OpenAI that communicates through natural language, processing user inputs, known as prompts, to produce textual responses. It can be used as an assisting tool for scholars in the process of editing, annotating, and curating (historical) texts. The authors evaluate ChatGPT primarily as practitioner-users, i.e., as editors working with the web-based graphical user interface, rather than as developers engaging with the underlying models or APIs. All experiments were performed using the ChatGPT web interface, specifically the GPT-4o model, accessed between July 2024 and July 2025 under both free and Plus tier access. ChatGPT is offered in multiple subscription tiers and model variants, each with different capabilities, usage limits, and available tools such as web browsing, image generation, and data analysis. During the revision period for this article, the default models and available features changed notably across tiers, illustrating the platform's rapid evolution. The source material draws on digital edition projects in which the authors are involved. Given the pace of development in both models and interface features, this review represents a time-bound snapshot rather than a comprehensive assessment.

3 In conceptualizing how generative AI might support scholarly editing, the Institute for Documentology and Scholarly Editing (IDE) has proposed a processing pipeline framework (Pollin et al. 2025). This model distinguishes between source material, edition as data, and edition as publication, connected through intermediate editorial steps. Through digitization, transcription, and annotation, the workflow progresses from raw source materials toward a critical text prepared for publication. The framework further differentiates between structural tasks (e.g., planning, documentation, tool integration, quality control) and operational tasks (e.g., text recognition, entity recognition, normalization, enrichment). This framework helps locate the stages at which

tools like ChatGPT may be effectively integrated into editorial workflows, and where their limitations become apparent.

4 ChatGPT is developed by OpenAI, a company founded in 2015 in San Francisco by a group of machine learning experts, research engineers, and scientists.[3] There is a huge interdisciplinary team of developers behind the system: the GPT-4 technical report lists 280 co-authors (OpenAI 2024). Financial support is provided by individuals and companies. In an interview with Bill Gates, OpenAI's CEO, Sam Altman has given an outlook on future developments: The focus will be on multimodality, reasoning ability, reliability, customizability, and personalization (Gates 2024, Chudleigh 2024).

5 From a tool-identification perspective, ChatGPT is accessed via a web-based interface in a modern browser. The system does not provide a stable version identifier or persistent identifiers for individual interactions. Although conversations can be shared via generated links, these links are not designed as durable scholarly references. This lack of stable versioning and citability constitutes a significant usability constraint for digital scholarly editing, where traceability and documentation of editorial decisions are essential.

# Methods and Implementation

6 The technical architecture of ChatGPT relies on generative pre-trained transformer (GPT) models, which are fine-tuned for natural conversation using a combination of supervised learning and reinforcement learning from human feedback. These LLMs belong to the broader class of foundation models, which are designed to support multiple downstream applications rather than being optimized for a particular, narrowly defined task. Therefore, ChatGPT can in principle be applied to a wide range of tasks at different stages of a digital scholarly editing workflow.

## Training Data, Transparency, and Reproducibility

7 The exact training data of the underlying models is not publicly disclosed, and the models themselves are not openly available. According to OpenAI, training data consists of "information that is publicly available on the internet, information that we license from third parties, and information that our users or our human trainers provide."[4] This limited transparency has been widely criticized, particularly in scholarly contexts where the provenance of training data is an important concern. Another significant limitation for

scholarly work relates to reproducibility. Even when identical prompts are used, ChatGPT does not reliably produce identical outputs across multiple runs. This behavior reflects the inherent non-deterministic nature of LLMs, which predict token probabilities in context based on the training data (Ouyang et al. 2023). As a result, identical prompts may produce slightly different outputs across multiple runs, even when the underlying model version remains unchanged, complicating the documentation, verification, and long-term traceability of editorial decisions.

8 Another factor affecting reproducibility concerns the availability of different model versions within the ChatGPT platform. As of June 2025, free-tier users have access by default to GPT-4o, OpenAI's flagship multimodal model, though access is subject to usage limits. Once these limits are reached, usage automatically defaults to a smaller, cost-efficient fallback. Users of subscription tiers such as Plus, Pro, or Team receive higher usage limits and access to a wider range of models through the model selector, including models optimized for complex reasoning tasks, high-performance multimodal interaction, and lightweight, cost-efficient processing. Because different models may produce different outputs for the same prompt, the presence of multiple model versions within a single platform introduces an additional source of variability when integrating ChatGPT into scholarly workflows.

## Interaction Model and Prompting

9 ChatGPT operates through a conversational interface in which users interact with the system using natural language prompts. While text is the primary input modality, interactions can also begin with images or voice. A prompt is the instruction or query provided to the model ranging from simple questions to complex scenarios, directing the LLM in its response generation. In addition to user prompts, ChatGPT also employs system prompts—internal instructions defined by the platform that guide the model's behavior and are not visible to the user. At the same time, users can influence the model's responses through personalization settings available in the ChatGPT interface via the user profile menu. Prompt engineering refers to the practice of designing and refining prompts to communicate the intentions of users to a large language model. Considerations of prompt length and structure are also relevant, as model inputs are processed as tokens and subject to context window limits. There are various prompting techniques, such as zero-shot, few-shot, and chain-of-thought prompting, each designed to approach the model differently based on the complexity of the task (Saravia 2022). In

editorial contexts, careful prompting is essential for achieving consistent results, particularly when tasks involve formalized output formats or domain-specific conventions.

10 All inputs and outputs processed by ChatGPT are internally segmented into tokens, which constitute the basic unit for both text processing and usage accounting.

> Tokens are the building blocks of text that OpenAI models process. They can be as short as a single character or as long as a full word, depending on the language and context. Spaces, punctuation, and partial words all contribute to token counts. This is how the API internally segments your text before generating a response.[5]

11 How exactly tokens are calculated is often not immediately intuitive for users. OpenAI suggests that as a rule of thumb for English texts one token corresponds to approximately four characters, or about three-quarters of a word. Consequently, 100 tokens amount to roughly 75 English words. OpenAI provides an online tokenizer that allows users to test how specific inputs are segmented into tokens.[6] While this approximation is slightly vague, an additional complication is that tokenization is language-dependent. This is especially relevant for digital scholarly editing, which very often involves historical texts in languages other than contemporary English.

## Configuration and Personalization

12 Beyond prompt-level interaction, ChatGPT also allows users to configure model behavior through customizable applications known as Custom GPTs. For creating them, no programming knowledge is necessary. The configuration screen allows you to enter a picture, a name of the GPT, a short description, detailed instructions, a conversation starter to facilitate the start of a conversation, and the upload of an extra knowledge base. When such files are provided, the system can retrieve relevant passages from them during response generation, thereby implementing a form of retrieval-augmented generation (RAG). It is recommended to provide the additional knowledge in a minimalistic format (e.g., JSON or Markdown) to save tokens instead of ‘chatty’ formats like XML, where each tag consumes tokens. While this built-in retrieval mechanism is suitable for small-scale text documents, larger corpora typically require interfacing with external databases.[7]

13 ChatGPT can be individually configured using ‘custom instructions’, a personalization feature distinct from Custom GPTs, within the personalization settings. Users may provide details about their activities, interests, or working context and indicate how the model should respond, for example by specifying a preferred tone or level of detail. These instructions persist across conversations so that users do not have to repeat these preferences in each prompt.

## Input, Output, and Data Analysis Capabilities

14 ChatGPT can handle and process text in various data formats like plain text, XML, HTML, JSON, CSV, or Markdown, as well as image, video, and audio as input. Likewise, it can generate output in most textual data formats mentioned; in addition, recent multimodal versions support audio output, while the generation of standalone video files is not natively supported in the standard ChatGPT interface. Advanced data analysis (ADA), integrated into GPT-4-class models, allows the model to generate Python code for performing exploratory data analysis tasks. It processes various structured input formats (e.g., Excel, CSV, PDF, and JSON) and produces corresponding outputs (e.g., Python Code, Visualizations, Charts, and interactive tables).

15 For developers, OpenAI provides an easy way to use its API via the programming language Python, the Curl command line tool for sending HTTP requests to APIs, and Node.js, a JavaScript framework to build rich applications, provided one has an API key. However, the focus of this article is the chatbot, and we will not explore the API.

## Positioning within the LLM landscape

16 ChatGPT exists within a broader and rapidly evolving ecosystem of generative language models. Proprietary systems include Claude by Anthropic, Gemini by Google, as well as models integrated into platforms like Microsoft Copilot. At the same time, open-weight and open-source models are advancing rapidly and in some benchmarks are even outperforming proprietary systems in certain tasks (Chen et al. 2024). Leading examples include the Llama family developed by Meta, Mistral AI’s Mistral and Mixtral models, and Alibaba’s Qwen series.

17 When assessing a language model's openness, it is important to differentiate between 'open-source models' and 'open-weight models'. Open source refers to sharing the full source code, the underlying algorithms, training methods, and data, while open weight models only share the pretrained model parameters, which allows use but limits transparency and control. Open-weight models are easier to use but less transparent, while open-source promotes transparency and allows modification, but also requires more developer effort (Ramlochan 2023). For digital scholarly editing, these distinctions are relevant for sustainability, inspectability, and long-term reuse. The current leading open-source models can be consulted at the Chatbot Arena[8].

## Examples and Current Uses of Generative AI in Digital Scholarly Editing

18 GPT models are generally considered not to be trained on specific domains or tasks, although user-guided fine-tuning can be applied to that end. They are general-purpose models that can assist in different tasks related to the digital editing pipeline, from data creation to final publication.

19 One area of experiments includes custom GPTs, i.e., user-configured applications built on top of a pre-trained GPT model. Examples include *teiCrafter*, an "expert automaton for converting plain text to TEI XML" (Pollin 2024); *TEI Scholar* (Jankovic); and *TEI Lex 0 Converter* (Miyagawa). Designing custom GPTs does not require training a new language model, but rather supplying detailed instructions and additional domain-specific knowledge, ideally in compact data formats such as Markdown or JSON to optimize token usage. Custom GPTs can be kept for internal use only or shared with the broader community; they remain bound to OpenAI's infrastructure and do not constitute independent editorial tools.

20 Beyond individual tools, broader community efforts further illustrate these developments. A workshop at the DHd 2024 conference in Passau, organized by the Institute for Documentology and Scholarly Editing (IDE), provided a broad overview of ongoing experiments with generative language models in digital scholarly editing (Pollin et al. 2024). The ten contributions covered a wide range of editorial tasks, including transcription, OCR post-correction, markup generation, normalization, named entity recognition, annotation, translation, and text summarization. As demonstrated in Pollin's workshop series, generative AI can be applied across diverse DH use-cases: from transforming unstructured data into TEI to generating and modeling RDF data to data

analysis and visualization (Pollin 2025). Alongside these exploratory applications, recent research has emphasized the need for robust pipelines and systematic evaluation in LLM-assisted scholarly workflows. Recent contributions assess, for example, the quality and reliability of LLM-assisted TEI encoding (Strutz and Vogeler 2025); Strutz forthcoming), the automated handling and resolution of editorial interventions (Strutz and Scholger 2025), and the automated extraction and normalization of metadata from historical letter transcriptions (Strutz and Scholger 2026). Galka and Vogeler compare the integration of different LLMs within a workflow for adding named entity recognition annotations (2026). These studies highlight both the potential of LLMs for complex, context-sensitive tasks and the methodological challenges that remain.

## Application for digital scholarly editing

21 Conversations with ChatGPT are stored on OpenAI's servers and can be accessed by users via the history option in the left sidebar. The conversations can be shared via generated links, allowing limited collaboration or documentation of interactions.[9] However, due to the non-deterministic nature of the models (discussed in the Introduction) and the absence of stable versioning, such records do not guarantee reproducibility in a scholarly sense. While non-reproducibility poses challenges for academic rigor, this behavior can also be used by editors as an advantage, namely by generating multiple versions of the same output (e.g., requesting several TEI encodings of a letter) and subsequently critically identifying the optimal encoding approach.

22 This approach might be viewed as a form of digital hermeneutics, potentially transforming what appears to be a limitation into a tool for deeper interpretive engagement with editorial decisions. ChatGPT can support editors with various tasks during the creation of a digital scholarly edition, ranging from summaries and translations to automated XML/TEI tagging and other NLP tasks. It can also be used as a 'buddy' to provide feedback on an existing data model or XML/TEI code. ChatGPT can even assist in writing small programs for the web presentation or data analysis of an edition, utilizing *Promptotyping*, a prompt engineering method on LLM-based software development.[10] From a utility perspective, ChatGPT primarily supports exploratory and preparatory stages of editorial work rather than final, authoritative decisions.

23 To evaluate ChatGPT's practical value for concrete editorial tasks, we selected two common use cases: (1) converting plain text into TEI-compliant XML and (2) identifying and classifying named entities. All experiments were performed using GPT-4o

in the browser interface. For more elaborate examples, we refer to slides including chat protocols in Pollin 2025.

24 In the first example, a letter from the digital edition *Joseph von Hammer-Purgstall. Korrespondenz* (Höflechner 2025 et al.) was provided as plain text and should be encoded following the *Guidelines of the TEI* (TEI Consortium 2026). The correspondence spans the late eighteenth to mid-nineteenth century and constitutes a multilingual scholarly exchange within the context of Oriental studies.

> Euer Gnaden,
> Unterzeichneter1 bittet um die gnädige Erlaubnis, künftigen Sonntag die Redoute2 besuchen zu dürfen3, weil er
> 1.) im letzten und sechsten Jahre seiner Studien und bereits ein Jahr länger an der Akademie ist als es seine Vorfahren waren, da ihnen diese Erlaubnis erteilt wurde.
> 2.) auf das ihm jederzeit von S[einer] H[ochwohlgeboren] dem Herrn Direktor erteilte Zeugnis seines guten Verhaltens und sittlichen Betragen stolz ist, und sich daher schmeichelt, dass die Besorgnis einiger daraus entstehenden üblen Folgen bei ihm nicht statt haben dürfte.
> 3.) die Vorschrift einer für diesen Fall geforderten anständigen und sicheren Begleitung nicht besser erfüllt haben könnte, indem er in der Gesellschaft der Frau Schwester des Herrn Direktors sich befinden wird.
> Den 20ten Hornung 1794
> Joseph Hammer
>
> 1.) PvTh D.2.1.14.
>
> 2.) Eine Tanzveranstaltung oder auch ein eleganter Maskenball. Um welche derartige Veranstaltung es sich hier handelt, ist nicht feststellbar.
>
> 3.) Die Erledigung durch Jenisch s. das nachfolgende Stück, findet sich auf demselben Blatt wie die vorangegangene Bitte HPs mit dem Vermerk „Von der Kanzlei den 20. Hornung 1794“.
>
> (Hammer-Purgstall an Jenisch 1794)

| | |
|---|---|
| As an expert in TEI encoding, please convert the following unstructured letter into well-formed TEI XML. Write the body of the TEI only. | Role<br>Task definition |
| Follow the encoding rules if applicable for the letter in question:<br>* enclose the letter text in a  element with the @type-attribute 'letter'.<br>* encode opener, closer, signatures, salutations, paragraphs and other textual structure;<br>* encode the notes as embedded elements instead of the end of the document;<br>* encode person names as <persName>;<br>* put the reference to norm data (e.g. GND or VIAF) into a @ref-attribute only if you are sure that it is the correct reference<br>* normalize the date using the @when attribute in <date><br>* resolve abbreviations that are marked with square brackets and use the <choice> mechanism with <abbr> and <expan>. Encode every word separately in an <choice> element, put the character(s) outside the square brackets into <abbr> and the character(s) within the square brackets into <expan>. Remove the square brackets.<br>* make sure that punctuation marks or articles are not part of salutations, dates, or person names. | Encoding rules |
| Output constraints:<br>* address only the steps that are in the instructions. do not add more.<br>* preserve the original text in its entirety<br>* produce well-formed TEI XML according to the TEI standard | Output constraints |
| Take a deep breath and work step by step. | Emotional prompting |

Fig. 1: Prompt instructions to convert a letter from Joseph von Hammer-Purgstall to Bernhard von Jenisch into TEI.

```xml
<body>
    <div type="letter">
        <opener>
            <salute>Euer Gnaden</salute>, </opener>
        <p>Unterzeichneter<note type="comment" n="1">PvTh D.2.1.14.</note> bittet um die gnädige
            Erlaubnis, künftigen Sonntag die <placeName>Redoute</placeName><note type="comment"
                n="2">Eine Tanzveranstaltung oder auch ein eleganter Maskenball. Um welche derartige
                Veranstaltung es sich hier handelt, ist nicht feststellbar.</note> besuchen zu
                dürfen<note type="comment" n="3">Die Erledigung durch Jenisch, siehe das
                nachfolgende Stück, findet sich auf demselben Blatt wie die vorangegangene Bitte HPs
                mit dem Vermerk „Von der Kanzlei den 20. Hornung 1794".</note>, weil er</p>
        <list>
            <item>
                <p>im letzten und sechsten Jahre seiner Studien und bereits ein Jahr länger an der
                    Akademie ist als es seine Vorfahren waren, da ihnen diese Erlaubnis erteilt
                    wurde.</p>
            </item>
            <item>
                <p>auf das ihm jederzeit von <choice><abbr>S</abbr><expan>einer</expan></choice>
                    <choice><abbr>H</abbr><expan>ochwohlgeboren</expan></choice> dem Herrn Direktor
                    erteilte Zeugnis seines guten Verhaltens und sittlichen Betragen stolz ist, und
                    sich daher schmeichelt, dass die Besorgnis einiger daraus entstehenden üblen
                    Folgen bei ihm nicht statt haben dürfte.</p>
            </item>
            <item>
                <p>die Vorschrift einer für diesen Fall geforderten anständigen und sicheren
                    Begleitung nicht besser erfüllt haben könnte, indem er in der Gesellschaft der
                    Frau Schwester des Herrn Direktors sich befinden wird.</p>
            </item>
        </list>
        <closer>
            <date when="1794-02-20">Den 20ten Hornung 1794</date>
            <signed>
                <persName>Joseph Hammer</persName>
            </signed>
        </closer>
    </div>
</body>
```

Fig. 2: XML/TEI encoding of the letter generated by ChatGPT.

25 The plain text of the letter together with the prompt (Fig. 1) was entered into the chat. In our case the prompt is structured in a task-oriented, rule-based format consisting of five main components: (1) *role framing*, which assigns the model the role of an expert in TEI encoding; (2) *task definition*, which specifies the transformation of an unstructured letter into well-formed XML/TEI; (3) *encoding rules*, which specify the selection ans use of TEI elements for structural and semantic components; (4) *output constraints*, which restricht the form of the response; and (5) *emotional prompting*, which encourages a step-by-step approach to improve reliability and consistency of the output. The resulting XML/TEI encoding Fig. 2 shows that the overall letter structure is correctly identified and appropriate XML/TEI elements are applied. Abbreviations are partially resolved, the date is normalized, and notes are embedded in the correct position in the text, even though the notes in the input text are placed at the end of the document. However, the recognition of personal names was less reliable; in addition, a place name was labeled, although only the tagging of person names was requested, suggesting that unwanted annotations should be explicitly excluded in the prompt. Potential risks include the incomplete preservation of the input text and the repositioning of structural elements such as dates or signatures in the output. Such rearrangements may suggest model-

internal assumptions about typical letter layouts, possibly reflecting generalized structural patterns learned during training. (Scholger, Strutz, and Pollin 2024)

26 In the second example, the same TEI annotation task was applied to reports of envoys present at the Imperial Diet of Regensburg of 1576 (Leeb et al. 2023). The envoys, Dr. Wolfgang Eulenbeck and Dr. Andreas Paulus, reported regularly to Prince-elector (Kurfürst) August von Sachsen, who was not present at the Imperial Diet. The form and structure of these reports are very similar to the above letter from the *Joseph von Hammer-Purgstall. Korrespondenz*. However, the significant difference is that they were written three centuries earlier in Early New High German. The correspondence between the envoys and August von Sachsen is published in the edition *Der Reichstag zu Regensburg 1576*.

27 The following quote shows the plain text of the letter from May 9 1576, to August von Sachsen:

> Neunden Maii, Regenspurgk rhäte. Präs. Annaburg, 12. Maii anno 76. Zu s. kfl. Gn. henden.
>
> Durchlauchtigster hochgeborner churfurst, ewern kfl. Gn. seint unsere underthenigste gehorsame dienste jeder zeit mit trewem vleiß zuvorn. Gnedigster herre.
>
> Auf euer kfl. Gn. gnedigsten bevelch und uns zugeschickter instruction haben wir uns aufn wegk nach Regensburgk erhoben, und seint gestern umb acht schlage vor mittage alhier ankommen. Und wiewol wir under wegs verstendigt, als solte die röm. ksl. Mt. auf den vierzehenden tag ditz monats Maii alhie angelangen, so werden wir doch durch des Reichs marschalch alhier berichtet, das sich irer ksl. Mt. aufbruch zu Wyen bis in mitten ermelts monats verweilett, und ire Mt. wenig vor ende desselben alhier ankommen werden, inmassen euer kfl. Gn. aus inligender copey der höchstgedachten ksl. Mt. schreiben an den bischoff [1v] alhier, item hern Conraden zu Pappenheim und doctor Viecheussern gnedigst haben zuersehen.
>
> So thuen euern kfl. Gn. wir hirbey underthenigst ein vorzeichnus ubersenden, was vor gesandten (derer doch wenig sein) bis auf itzo alhier angelangt.
>
> Und, euer kfl. Gn. mit underthenigstem gehorsamb trewe dienste zuleisten, seint wir schuldig und geflissen. Datum Regensburgk, den neunden Maii anno 76.
>
> Ewer kfl. Gn.
>
> underthenigste gehorsame

Wolffgang Eylenbeck, Dr.

Andreas Paull propria[?]

(Bericht: Dr. Wolfgang Eulenbeck und Dr. Andreas Paulus an Kurfürst August von Sachsen, 1576-05-09)

```
<body>
    <div type="letter">
        <opener>
            <dateline>
                <placeName>Regenspurgk</placeName>, <date when="1576-05-09">Neunden Maii</date>
            </dateline>
            <salute>Zu <choice><abbr>s.</abbr><expan>seiner</expan></choice>
                <choice><abbr>kfl.</abbr><expan>kurfürstlichen</expan></choice> Gn. henden</salute>
        </opener>
        <p> Durchlauchtigster hochgeborner churfurst,
                    <choice><abbr>ewern</abbr><expan>eweren</expan></choice>
            <choice><abbr>kfl.</abbr><expan>kurfürstlichen</expan></choice> Gn. seint unsere
                    <choice><abbr>underthenigste</abbr>
                <expan>unterthenigste</expan></choice> gehorsame dienste jeder zeit mit trewem
                    <choice><abbr>vleiß</abbr><expan>vleiß</expan></choice> zuvorn.
                <salute>Gnedigster herre</salute>. </p>
        <p>[ ... ]</p>
        <p> So thuen euern <choice><abbr>kfl.</abbr><expan>kurfürstlichen</expan></choice> Gn. wir
            hirbey <choice><abbr>underthenigst</abbr>
                <expan>unterthenigst</expan></choice> ein vorzeichnus ubersenden, was vor gesandten
            (derer doch wenig sein) bis auf itzo alhier angelangt. </p>
        <p> Und, euer <choice><abbr>kfl.</abbr><expan>kurfürstlichen</expan>
            </choice> Gn. mit <choice><abbr>underthenigstem</abbr>
                <expan>unterthenigstem</expan></choice> gehorsamb trewe dienste zuleisten, seint wir
            schuldig und geflissen. </p>
        <closer>
            <dateline>
                <placeName>Regensburgk</placeName>, <date when="1576-05-09">den neunden Maii anno
                    76</date>
            </dateline>
            <salute>Ewer <choice><abbr>kfl.</abbr><expan>kurfürstlichen</expan></choice>
                Gn.</salute>
            <signed>
                <persName>Wolffgang Eylenbeck</persName>,
                        <choice><abbr>Dr.</abbr><expan>Doctor</expan></choice>
            </signed>
            <signed>
                <persName>Andreas Paull</persName>
                <choice>
                    <abbr>propria[?]</abbr>
                    <expan>propria</expan>
                </choice>
            </signed>
        </closer>
    </div>
</body>
```

Fig. 3: The letter to August von Sachsen encoded in TEI using ChatGPT GPT-4o. The entire second paragraph was left out in order to fit the beginning and end on the picture.

28 The same prompt instructions as in Fig. 1 were used together with the plain text of the letter. However, the statement “keep the original language” had to be added to ensure that the source text was not normalized to standard German, which happened during the initial run. The result in Fig. 3 shows that, even with a historical language like Early New High German texts, ChatGPT identifies the structure of the document reasonably well.

29 Close observation of the result shows that some text is missing. For some reason, the part of the opener indicating when the letter was presented to August von Sachsen: ‘Präs. Annaburg, 12. Maii anno 76’ is missing. Furthermore, identification and expansion of abbreviations is a challenge. While abbreviations, such as ‘s.’, ‘Mt.’ and ‘kfl.’ are correctly resolved, others were not even identified as abbreviations, e.g., ‘Gn.’, or words were incorrectly identified as abbreviations, e.g., ‘ewern’, ‘bevelch’, ‘instruction’. This can easily be resolved by providing a list of the abbreviations used in editions of the Imperial Diets.[11] Dates are precisely tagged and correctly normalized using the `@when` attribute. However, ChatGPT does not automatically take the Gregorian calendar reform into consideration. It has to be explicitly mentioned to add `@calendar`.

Here’s a structured table of the people and places extracted from the TEI, with normalized names and authority references where identifiable:

| Original Name (EHND) | Type | Normalized Name | Authority Reference |
|---|---|---|---|
| Wolffgang Eylenbeck | Person | Wolfgang Eylenbeck | GND 104287020 (also VIAF 32430582) link.springer.com +4 deutsche-biogra... +6 |
| Andreas Paull | Person | Andreas Paull | – (no match found) |
| hern Conraden zu Pappenheim | Person | Conrad zu Pappenheim | – (insufficient data for authority identification) |
| doctor Viecheussern | Person | Dr. Viecheussern | – (no match found) |
| bischoff | Person (title) | Bischof (unspecified) | – (generic term, no specific person identified) |
| Regenspurgk / Regensburgk | Place | Regensburg | GND 4306129-0 |
| Wyen | Place | Wien (Vienna) | GND 4002339-4 |

**Notes:**

- **Wolffgang Eylenbeck** is confirmed via GND and VIAF IDs deutsche-biogra... +6 .
- **Regensburg** identified with GND and corroborated by historical descriptions en.wikipedia.org +10 .
- **Wien (Vienna)** is referenced historically as “Wyen,” matched with Vienna GND archive.org +2 .
- For **Andreas Paull**, **Conrad zu Pappenheim**, **Dr. V heussern**, and the generic **Bischof**, there were no clear authority matches—likely due to historical obscurity or alternative spellings.

Fig. 4: The figure shows the first result from ChatGPT trying to identify persons and places in the letter of 1576. Wrongly identified places are highlighted.

Go again through the list of people and places and only provide confirmed references to authority files. In case of the places use geonames.

Here are **only the confirmed entities** with authority file references (using GeoNames for places):

| Original Name (EHND) | Type | Normalized Name | Authority Reference |
|---|---|---|---|
| Regenspurgk / Regensburgk | Place | Regensburg | GeoNames 2849483 (Regensburg, Germany) geonames.nrcan... geonames.org +9 |
| Wyen | Place | Vienna | GeoNames 2761369 (Vienna, Austria) |

No personal names had confirmed authority file matches.

Sources

Fig. 5: Geoname references of the places generated by ChatGPT.

30 The tagging of people and places is also fine, but the identification and linking to authority files, as asked in the prompt, is missing. When asked to identify people and places and list them in a table with the columns (1. original name as found in the letter, 2. type of entity, 3. normalized name, 4. reference to an authority file, such as GND, VIAF or Wikidata) the result shows that at first, only one person could be correctly identified, Wolfgang Eylenbeck, 'GND 104287020 (also VIAF 32430582). ChatGPT also provides the link to sources for the identification: 'link.springer.com' and 'deutsche-biographie.de'. However, only the link to the *Deutsche Biographie* ('https://www.deutsche-biographie.de/pnd104287020.html') provides additional information about Eylenbeck and the references to the correct authority files. After being asked again with consideration of the date of the source, ChatGPT identifies 'Conrad zu Pappenheim' (Conrad von Pappenheim) correctly with GND and VIAF identifiers. The other references to people seem to be too vague. In the case of 'doctor Viecheussern', the Imperial Vice-Chancellor Sigismund Viehauser (oder Vieheuser), the very particular variance of the name could be the reason. It is also interesting that ChatGPT was not able to correctly identify the place names mentioned, although it recognizes that 'Regenspurgk' refers to 'Regensburg' and 'Wyen' to 'Wien (Vienna)'. The GND identifiers provided by ChatGPT for the two places do not actually exist (Fig. 4). However, if asked to provide Geoname references for the places, both are correctly identified (Fig. 5).

## Usability, sustainability, and maintainability

31 Generally, ChatGPT can be accessed via the OpenAI website, and an API is available for programmatic use. The tool requires minimal setup, and the web interface is simplistic and easy to use. It works well on all modern web browsers and devices, and a ChatGPT app is available for mobile devices. Users require internet access and an OpenAI account. Additional services beyond the free tier, including expanded API usage and processing capabilities, are only available under a pay plan based on OpenAI's token system.[12] We do not go into detail about pricing as it depends on the model used; however, it has to be noted that for more complex and longer tasks, it can get very expensive. This makes it less attractive for sustained scholarly work and creates unequal conditions, as not all academics or projects might have the resources to cover the costs. It is also worth noting that access to ChatGPT varies globally.[13] In regions where the tool is restricted, such as China, alternatives like DeepSeek and Ernie Bot are available (Yang 2023; Wikipedia contributors 2024). While these tools offer similar functionalities, their effectiveness for digital scholarly editing would require separate evaluation.

32 The various language models used within ChatGPT, and their differences, are briefly documented and compared on the OpenAI website.[14] OpenAI also has a help center and maintains a developer forum where, currently, mostly API-related issues are discussed. ChatGPT has a big user community and is widely used. Therefore, if you run into a general problem, the chance is very high that someone else already had a similar problem, and the solution can be found online. OpenAI also provides documentation and guidelines for the use of the tool. For instance, the guidelines for prompt engineering explain strategies and suggestions for crafting effective prompts to get better results from ChatGPT [15]. In addition to the material provided by OpenAI, it has to be pointed out that, due to the great interest and broad application of chatbots in recent years, numerous introductions and documentation about working with ChatGPT were developed by individuals, institutions, and organizations all over the world. This also includes some subject-specific introductions for Digital Humanities and digital scholarly editing (e.g. Pollin 2025).

33 While interacting with ChatGPT is quite easy, creating effective prompts requires some training. Prompt engineering has become its own discipline, and mastering these techniques is particularly important in digital humanities contexts, where specialized

terminology, historical language material, and encoding standards require precise instructions (Saravia 2022, Pollin 2025). Well-designed prompts lead to better results and reduce frustration in the use of ChatGPT. It has to be emphasized that the use of ChatGPT is very much experimental in most disciplines, and its full capacities are not yet clear; further study is necessary. This holds especially true for digital humanities and its use for digital scholarly editing.

34 ChatGPT is not open source. The tool itself and its development are closed off. This means that neither the source code nor the underlying models can be reused or built upon. However, the performance of GPT has been analyzed extensively since its release. Users all over the world from various disciplines have experimented and published benchmark tests and evaluations.[16] This lack of openness raises concerns with regard to transparency, sustainability, and long-term reusability in scholarly workflows.

35 OpenAI assures its users that all data shared with ChatGPT is encrypted, and the input and the output you receive from ChatGPT are rightfully yours.[17] Nevertheless, there are concerns that OpenAI uses input data for training purposes and that sensitive information may end up in the company's database and become available to other users of the tool. Another concern that has been voiced is that OpenAI collects substantial data about its users and their behavior (Merton-McCann 2023).

36 Finally, the question of citation must be addressed. If ChatGPT is used in research or for academic writing, its involvement needs to be acknowledged. This is also an issue when we want to encourage our students to use it as a tool. Conversations are tightly bound to a user account, and there are no permalinks, no DOI, no URL that allows for persistent sharing. Consequently, referencing ChatGPT is a big issue for students and researchers. There is not yet a general consensus on how to cite ChatGPT conversations.[18] However, ChatGPT is usually not considered an author. This also means that ChatGPT – or any AI – cannot claim copyright and any AI-generated output is automatically considered public domain (Scholger 2025). APA, for example, suggests that 'chats' with the AI cannot be cited as personal communications as there is no person communicating.[19] It is typically recommended to include the prompt or the entire conversation for reference (for instance, in an appendix).[20] This reminds us of the recommendations from the early 2000s to print out entire websites and add them to an essay. However, this approach can quickly become unwieldy, resulting in an

overwhelming flood of appendices, as conversations with ChatGPT can be quite extensive.

## User interaction, accessibility, and bias

37 ChatGPT is a chatbot, and in this review, we focus on the GUI provided by OpenAI at https://chat.openai.com/. The interface requires a user account and login and is designed primarily for conversational interaction, i.e., asking questions and receiving responses. Details have already been described in previous sections. From a usability perspective, the interface is minimalistic and straightforward, supporting quick access to the core functionality of the tool: the conversation. One gets the impression that OpenAI did not invest too much effort into the design, as their interest and attention are on other aspects. And indeed, critical remarks can be found regarding its simplicity and limited customization options (e.g., Sebastian 2023).

38 ChatGPT is described as a multilingual chatbot that can communicate in over 50 languages.[21] It also provides the possibility to translate from one language into another, making it a competitor for Google Translate, DeepL, and similar services (see, for instance, Quillen 2023; PCMAG 2023). Also, the speech-to-text Audio API based on the Wisper model, which is also part of the ChatGPT ecosystem, provides support for many languages. It was trained on 98 languages, but for only about 58 languages, the word error rate is lower than 50%.[22]

39 ChatGPT is also frequently discussed in the context of accessibility and the provision of barrier-free content. The OpenAI web interface is simplistic and screen-reader friendly. But the actual benefit comes from the use of the ChatGPT API in various new applications. For instance, the description of images is one such use-case[23].

40 At the end of this examination of ChatGPT's usability, it is necessary to address the question of bias. Even if we 'talk' to a machine, ChatGPT is not objective. The chatbot is trained on an enormous amount of data, most of it from a North American and European context, and the biases introduced through that data influence the response of the language model. Gender, racial, religious, age, or political bias, and other negative behaviors have been correctly criticized in the past.[24] If you ask ChatGPT, it will explain to you that it tries to be objective, but it is not free of bias. OpenAI provides a disclaimer about bias and stereotypes in ChatGPT responses on its website. They suggest that users should be careful with generated content, particularly when dealing with culturally

sensitive or historically situated material.[25] Another aspect that is currently discussed emotionally in politics and the media concerns the ecological dimensions of using AI, especially the increase in power consumption of data centers and companies in the coming years. Sustainability is therefore an important topic that generates more and more attention and forms part of the broader context in which decisions about adopting such tools for scholarly editing are made (Rohde et al. 2024).

## Conclusion

41 ChatGPT is not a tool that was explicitly developed with the digital humanities or the scholarly editing community in mind. Nevertheless, it has a wide range of potential applications for researchers and editors and can support different stages of the development or analysis of digital editions. It can assist with transcription and data creation, support the development of scripts for data processing and analysis, and facilitate related tasks.

42 There is still relatively little research on the use of LLMs for editorial purposes, and the possibilities of what can be done with ChatGPT and similar AI tools are constantly changing. We are aware that the evaluation of such technology is only a snapshot that might not be relevant in a gcouple of months, and that all predictions for future developments are quite difficult as development moves so quickly. As we outlined in the article, we chose to evaluate ChatGPT (GPT-4o) over other LLMs as it is well-known and currently widely used by humanities scholars.

43 Several limitations significantly constrain the use of ChatGPT as a tool for scholarly editing. First, the development of the tool is closed off as it is a commercial product. For editors, this results in a black-box situation that can be vaguely controlled using prompts. If prompts are not carefully crafted, the results can be misleading or even totally wrong. Often, negative feedback about ChatGPT is the result of knowing too little about its capabilities and how to create effective prompts. Second, it remains unclear how OpenAI handles the data provided by users and to what extent the data is used to train future models. Therefore, users have been warned to be careful and not to share sensitive or unpublished material with AI. For scholarly editing, this significantly restricts the range of sources for which ChatGPT can be used responsibly. Third, while the chatbot interface is suitable for small-scale experiments and exploratory tasks, complex multi-stage editorial workflows for an entire corpus - from the TEI encoding to the recognition of named entities and their enrichment with norm data - can only be

effectively implemented via the API. The associated costs cannot be covered by every project and also contradict the principles of an open-knowledge society and equal access to research. This is why it is so important that we focus more on developments in the area of open-source LLMs in the future. Finally, ethical issues caused by the bias in the models, along with ecological dimensions of using AI, such as high-power consumption, are substantial issues and have to be kept in mind when incorporating these tools into scholarly workflows.

44 The use cases presented in this review demonstrate ChatGPT's high potential as an assistant for digital editing. Tasks that previously required several tools and technical expertise can now be performed with comparatively little effort using carefully designed prompting instructions. At the same time, we have also discussed that the quality of the results can diverge depending on the source material of the edition and a user's ability to craft 'good prompts'. In any case, a human editor must remain in the loop to control, evaluate, and, if necessary, further process the output. This is especially highlighted by our little experiment using German texts from two different historical language periods. It shows that ChatGPT works less effectively with historical languages.

## Notes

1. LLMs are trained with deep-learning algorithms on vast corpora of web pages, books and articles to learn the patterns and structures of language. With multi-billion-parameter architectures, they are able to generate context-aware text.

2.

Martina Scholger and Roman Bleier are both affiliated with the Department of Digital Humanities at the University of Graz, where they work in the field of digital scholarly editing — Scholger as a senior scientist and Bleier as a postdoctoral researcher. We thank Christopher Pollin of DH Craft for reading an early draft of this article and for providing valuable feedback and helpful suggestions, and Thea Schaaf for proofreading.

3. Brockman, Sutskever, and OpenAI 2024, here: https://web.archive.org/web/20241007222900/https://openai.com/blog/introducing-openai/.

4. OpenAI 2025, here: https://web.archive.org/web/20240913163752/https://help.openai.com/en/articles/7842364-how-chatgpt-and-our-language-models-are-developed.

5. OpenAI 2025, here: https://web.archive.org/web/20251012012828/https://help.openai.com/en/articles/4936856-what-are-tokens-and-how-to-count-them.

6. https://web.archive.org/web/20251008185403/https://platform.openai.com/tokenizer/.

7. Saravia 2022, here https://web.archive.org/web/20250808170700/https://www.promptingguide.ai/techniques/rag.

8. An overview of leading chatbots is provided by the 'Chatbot Arena' (Chiang et al. 2024), an online platform that ranks LLMs based on crowdsourced human preference evaluations.

9. OpenAI 2025, here: https://web.archive.org/web/20251118195451/https://help.openai.com/en/articles/7925741-chatgpt-shared-links-faq.

10. https://web.archive.org/web/20260601000000*/https://lisa.gerda-henkel-stiftung.de/digitale_geschichte_pollin.

11. Such a list is provided by every edition of the Imperial Diets: e.g., https://gams.uni-graz.at/o:rta1576.bt17331b5u.

12. OpenAI uses a token system for input, output, and training usage. https://web.archive.org/web/20260314111058/https://chatgpt.com/pricing/. But this constantly changes.

13. For a brief period of time in 2023 even in Italy it was banned due to issues with the data protection regulations, DigitalTrends: https://web.archive.org/web/20260316110636/https://www.digitaltrends.com/computing/these-countries-chatgpt-banned/. See also a recent article (June 2024) about China: https://web.archive.org/web/20240630064038/https://edition.cnn.com/2024/06/21/tech/apple-ai-chatgpt-ban-china/index.html.

14. https://web.archive.org/web/20260317093543/https://developers.openai.com/api/docs/models.

15. https://web.archive.org/web/20260308121716/https://developers.openai.com/api/docs/guides/prompt-engineering

16. For instance, the ability of ChatGPT 3.5 and GPT 4 to write correct code was conducted by Liu et al. 2023 and a test on various NLP tasks was conducted by Laskar et al. 2023.

17. https://web.archive.org/web/20260312100111/https://openai.com/enterprise-privacy/.

18. As an example we provide here the recommendation by the University of Graz, last update 18.08.2025: https://lehren-und-lernen-mit-ki.uni-graz.at/de/ki-nutzung-kennzeichnen/.

19. 'Unfortunately, the results of a ChatGPT "chat" are not retrievable by other readers, and although nonretrievable data or quotations in APA Style papers are usually cited as personal communications, with ChatGPT-generated text there is no person communicating.' https://apastyle.apa.org/blog/how-to-cite-chatgpt.

20. https://apastyle.apa.org/blog/how-to-cite-chatgpt; APA policy on the use of generative artificial intelligence (AI) in scholarly materials, in: *APA Publishing Policies (Aug. 2023)*: https://web.archive.org/web/20260301102507/https://www.apa.org/pubs/journals/resources/publishing-policies?tab=3.

21. The chatbot and its user interface are constantly evolving and updated, but that is the situation in July 2024. OpenAI: How to use ChatGPT in a language other than English (alpha), https://help.openai.com/en/articles/8357869-how-to-use-chatgpt-in-a-language-other-than-english-alpha.

22. OpenAI: Speech to text: https://platform.openai.com/docs/guides/speech-to-text; https://platform.openai.com/docs/guides/speech-to-text/supported-languages.

23. https://web.archive.org/web/20260311135431/https://netzpolitik.org/2023/gpt-4-das-naechste-grosse-ding-fuer-digitale-zugaenglichkeit/.

24. See for instance examples on the UNESCO website (UNESCO 2023); on the matter see also the end of the keynote lecture by Pierazzo (2024).

25. OpenAI 2025 "Is ChatGPT biased?", https://web.archive.org/web/20260311133649/https://help.openai.com/en/articles/8313359-is-chatgpt-biased.

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# Factsheet

| Resource reviewed | |
|---|---|
| **Title** | ChatGPT |
| **Editors** | OpenAI |
| **URI** | https://openai.com/ |
| **Publication Date** | 2022 |
| **Date of last access** | 01.02.2026 |

| Reviewer | |
|---|---|
| **Name** | Scholger, Martina |
| **Affiliation** | University of Graz |
| **Place** | Graz, Austria |
| **Email** | martina.scholger (at) uni-graz.at |

| Reviewer | |
|---|---|
| **Name** | Bleier, Roman |
| **Affiliation** | University of Graz |
| **Place** | Graz, Austria |
| **Email** | roman.bleier (at) uni-graz.at |

| General information | | |
|---|---|---|
| **Software type** | What type of software is it? (cf. Catalogue 0.1.1) | Software tool |
| **Identification of the environment** | On which platform runs the tool? (cf. Catalogue 1.4) | Web browser |
| **Purpose** | For what purpose was the tool developed? (cf. Catalogue 1.5) | developed to accomplish a general task |
| **Funding** | Which is the financial model of the tool? (cf. Catalogue 1.6) | Free/open |

| | | |
|---|---|---|
| **Maturity** | What is the development stage of the tool? (cf. Catalogue 1.5) | Release |
| **Methods and implementation** | | |
| **Programming Language** | Which programming languages and technologies are used? (cf. Catalogue 2.3) | Other: large language model |
| **Reuse** | Does the tool reuse portions of other existing software? (cf. Catalogue 2.3) | no |
| **Input format** | Which input formats are supported? (cf. Catalogue 2.4) | .xml, .xml/ tei, .txt, .pdf, .csv, .html, .json, .rdf |
| **Output format** | Which output formats are supported? (cf. Catalogue 2.4) | .xml, .xml/tei, .txt, .csv, .html, .json, .rdf |
| **Encoding** | Which character encoding formats are supported? (cf. Catalogue 2.4) | latin-1, utf-8, utf-16, Unknown |
| **Encoding preprocessing** | Is a pre-processing conversion included? | yes |
| **Dependencies** | Does the documentation list dependencies on other software, libraries or hardware? (cf. Catalogue 3.2) | no |
| **Dependencies installation** | If yes, is the software handling the installation of dependencies during the general installation process (you don't have to install them manually before the installation)? | no |
| **Documentation and support** | | |
| **Documentation** | Is documentation and/or a manual available? (tool website, wiki, blog, documentation, or tutorial) (cf. Catalogue 3.4) | yes |
| **Documentation format** | Which format has the documentation? (cf. Catalogue 3.3) | .html |
| **Documentation parts** | Which of the following sections does the documentation contain? (cf. Catalogue 3.3) | Step-by-step instructions, Examples, FAQ, API documentation (if relevant) |

| | | |
|---|---|---|
| **Documentation language** | In what languages is the documentation available? (cf. Catalogue 3.3) | Arabic, Chinese, English, French, German, Italian, Japanese, Portuguese, Russian, Spanish |
| **Support** | Is there a method to get active support from the developer(s) or from the community? (cf. Catalogue 3.4) | yes |
| **From of support** | Which form of support is offered? (cf. Catalogue 3.4) | Forum, Other: e-learning and online training |
| **Issue tracker** | Is it possible to post bugs or issue using issue tracker mechanisms? (cf. Catalogue 3.4) | no |
| **Usability and sustainability** | | |
| **Build and install** | Grade how straightforward it is to build or install the tool on a supported platform: (cf. Catalogue 3.6) | straightforward |
| **Tests** | Is there a test suite, covering the core functionality in order to check that the tool has been correctly built or installed? (cf. Catalogue 3.7) | no |
| **Portability and interoperability** | On which platforms can the tool/ software be deployed? (cf. Catalogue 3.8) | Not applicable (if web-based for example) |
| **Devices** | On which devices can the tool/ software be deployed? (cf. Catalogue 3.8) | Not applicable (if web-based for example) |
| **Browsers** | If the tool is web-based: On which browsers can the tool/ software be deployed? (cf. Catalogue 3.8) | Mozilla Firefox, Google Chrome, Safari |
| **Plugins** | If the tool is web-based: Does the tool rely on browser plugins? (cf. Catalogue 3.8) | no |
| **API** | Is there an API for the tool? (cf. Catalogue 3.8) | yes |
| **Code** | Is the source code open? (cf. Catalogue 3.9) | no |
| **License** | Under what license is the tool released? (cf. Catalogue 3.9) | No explicit license / all rights reserved |

| | | |
|---|---|---|
| **Credits** | Does the software make adequate acknowledgement and credit to the project contributors? (cf. Catalogue 3.9) | no |
| **Registered** | Is the tool/software registered in a software repository? (cf. Catalogue 3.9) | no |
| **Possible contribution** | If yes, can you contribute to the software development via the repository/development platform? | not applicable |
| **Analysability, extensibility, reusability of the code** | | |
| **Analysability** | Can the code be analyzed easily (is it structured, commented, following standards)? (cf. Catalogue 3.10) | no |
| **Extensibility** | Can the code be extended easily (because there are contribution mechanisms, attribution for changes and backward compatibility)? (cf. Catalogue 3.10) | not applicable |
| **Reusability** | Can the code be reused easily in other contexts (because there are appropriate interfaces and/or a modular architecture)? (cf. Catalogue 3.10) | not applicable |
| **Security and privacy** | Does the software provide sufficient information about the treatment of the data entered by the users? (cf. Catalogue 3.11) | no |
| **Supportability and maintenance** | Is there information available whether the tool will be supported currently and in the future? (cf. Catalogue 3.12) | no |
| **Citability** | Does the tool supply citation guidelines (e.g. using the Citation File Format)? (cf. Catalogue 3.13) | no |
| **User interaction, GUI and visualization** | | |
| **User profile** | What kind of users are expected? (cf. Catalogue 4.1) | General public |

| | | |
|---|---|---|
| **User interaction** | What kind of user interactions are expected?<br>(cf. Catalogue 4.1) | Reading, Text editing, Text analysis, Searching, Comparing |
| **User Interface** | What kind of interface does the tool provide?<br>(cf. Catalogue 4.2 and 0.1.1) | Graphical User Interface (GUI) |
| **Visualization** | Does the tool provide a particular visualizations (in terms of analysis) of the input and/or the output data?<br>(cf. Catalogue 4.3) | no |
| **User empowerment** | Is the user allowed to customize the functioning of the tool and the output configuration?<br>(cf. Catalogue 4.4) | yes |
| **Accessibility** | Does the tool provide particular features for improving accessibility, allowing „people with the widest range of characteristics and capabilities" to use it?<br>(cf. Catalogue 4.5) | yes |